

Title: PINN-Aided Gravity Anomaly Map Matching for GNSS-Denied Inertial Navigation

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Motivation

Reliable navigation in GNSS-denied environments is critical for autonomous systems and defense applications. Low-fidelity gravity models fail to capture local anomalies that carry precise positional information, and the differentiable nature of PINNs makes them naturally compatible with EKF-based state estimation – eliminating the need for manual linearization and making the overall pipeline both principled and practical.

Objective

This project aims to correct INS drift in GNSS-denied environments by training a Physics-Informed Neural Network (PINN) to learn the residual gravity anomaly (δg). The PINN-predicted anomaly is compared against onboard gravimeter measurements to generate an innovation term in an EKF, where the required Jacobian is computed via automatic differentiation of the PINN.

Deliverable

We deliver a working simulation of the proposed PINN-assisted INS-EKF navigation system for GNSS-denied environments. The system will include a trained PINN gravity model that provides gravity predictions used to correct INS drift through the EKF. The results will be demonstrated using simulated GNSS-denied quadrotor trajectory with GGMplus gravity data, and evaluated via position RMSE comparison between raw IMU dead-reckoning and PINN+EKF correction.

Research Plan

1. Description of the Algorithm

The proposed algorithm corrects INS drift in GNSS-denied environments through the following. A PINN is trained to learn the residual gravity anomaly, δg , given the position of a vehicle. The data of the PINN will be based on the GGMplus gravity data set that gives us gravity anomalies up to 200m resolution over areas of latitude and longitude. During navigation, the measured anomaly modeled as, $\delta g_{measured} = \delta g_{GGMplus} + n$, where n is sensor noise, is compared against the PINN-predicted anomaly to form an innovation term v , which is fed into an EKF to correct the INS navigation state. The measurement Jacobian $H = \nabla_x \delta g_{PINN}$ is computed via automatic differentiation, producing a corrected navigation state without GNSS.

2. Architecture

The proposed architecture integrates a PINN with an INS and EKF framework to provide an additional aiding source in a GNSS-denied environment. The PINN takes position coordinates (lat, long, alt) as input and outputs the corresponding gravity anomaly δg through a fully connected neural network with 4 hidden layers of 64 neurons each, Tanh activation functions, and skip connections. The EKF

operates in two steps: in the **predict step**, the INS propagates the navigation state $\hat{x}_{k|k-1}$ using IMU measurements:

$$\hat{x}_{k|k-1} = F\hat{x}_{k-1|k-1}, P_{k|k-1} = FP_{k-1|k-1}F^T + Q$$

In the **update step**, the PINN predicts $\delta g_{PINN}(\hat{x}_{k|k-1})$ at the predicted position to form the innovation:

$$\begin{aligned} v_k &= \delta g_{measured} - \delta g_{PINN}(\hat{x}_{k|k-1}) \\ K_k &= P_{k|k-1}H^T(HP_{k|k-1}H^T + R)^{-1} \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k v_k, P_{k|k} = (I - K_k H)P_{k|k-1} \end{aligned}$$

The Jacobian $H = \nabla_x \delta g_{PINN}$ is computed via automatic differentiation, eliminating manual linearization.

3. Loss Function

$$\begin{aligned} L_{total} &= L_{data} + \lambda L_{physics} \\ L_{data} &= \frac{1}{N_{data}} \sum_{i=1}^{N_{data}} (|\Delta g_{PINN}(r_i) - g_{true}(r_i)| + \frac{|\Delta g_{PINN}(r_i) - \Delta g_{true}(r_i)|}{|\Delta g_{true}(r_i)|}) \\ L_{physics} &= \frac{1}{N_{coloc}} \sum_{j=1}^{N_{coloc}} |\nabla^2(\Delta g_{PINN}(r_j))| \end{aligned}$$

The loss follows the PINN-GM-III structure. Both absolute and relative error terms are included to prevent low-altitude bias, as surface gravity magnitudes are much larger than those at altitude. The boundary condition is enforced via a weighted sum:

$$\Delta g(r) = \omega_{NN}(r)\Delta g_{NN}(r) + \omega_{BC}(r)\Delta g_{BC}(r)$$

4. Training

Training data consists of 32,768 position-gravity anomaly pairs synthetically generated from the GGMplus N30W120 tile over the USC campus region (34.02°N, 118.29°W), sampled on a 64x64 uniform grid at 8 altitude levels (0-200 m). Data is normalized using StandardScaler fitted to the local subgrid and split into training (22,937), validation (4,915), and testing (4,916) sets. The model is trained using the Adam (lr=2⁻⁸, batch size 512, $\lambda=10^{-3}$). Later on in a different test the model was retested on the GGMplus data with an altitude correction that was not a part of the original data since the GGMplus only provides surface level data.

5. Validation

Hyperparameters (hidden layers, neurons, learning rate λ) were selected by evaluating δg prediction error on a held-out validation set, with the Jacobian monitored to ensure well-conditioned EKF linearization. The configuration with lowest validation MSE was selected. And other parameters such as learning rate were based off of the PINN-GM-III paper as a good working baseline.

6. Testing

The pipeline will be tested on a simulated GNSS-denied quadrotor trajectory generated using a 6-DOF dynamics model with RK4 integration, with realistic IMU noise and bias ($\sigma_{gyro}=0.002$ rad/s, $\sigma_{accel}=0.05$ m/s²). Performance was evaluated by comparing position RMSE between: (1) raw INS/IMU dead reckoning only. (2) PINN-learned anomaly map with EKF correction. GGMplus gravity data was used as the ground truth gravity reference, with a simulated gravimeter noise of $\sigma = 0.5$ mGal.

Results and Discussion

PINN Training Results

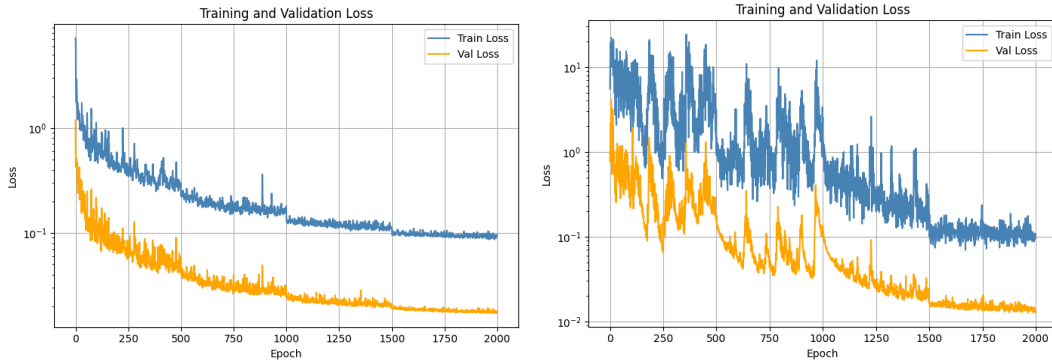


Figure 1: Training and Validation Loss Curves (left - Test 1, right - Test 2)

Two PINN implementations were evaluated to understand the effect of normalization scope. Test 1 fits StandardScaler fitted to the local USC subgrid (mean = -75.1 mGal, std = 8.1 mGal), correctly capturing the operating region's gravity distribution. Test 2 fits the scaler to the full GGMplus tile range (mean = -105.9 mGal, std = 21.8 mGal), introducing a distribution mismatch with the test region. Both models share identical architecture and training settings as described in Research Plan - Training.

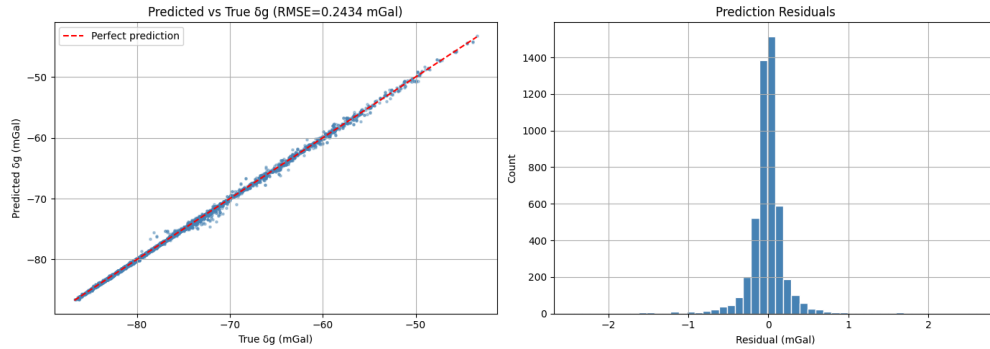


Figure 2: Predicted vs True δg scatter plot + Residuals Histogram (Test 1)

[Table 1: Test Set Performance of Both Models]

Model	RMSE (mGal)	MAE (mGal)	Relative Error	USC δg Prediction	USC Error
Test 2	0.5300	0.2821	0.31%	-95.39 mGal	16.27 mGal

Test 1	0.2434	0.1383	0.20%	-79.30 mGal	0.19 mGal
PINN-GM-III (paper benchmark)	-	-	0.30%	-	-

Test 1 outperforms both Test 2 and the PINN-GM-III benchmark of 0.30% reported by Martin & Schaub. The critical difference is the normalization scope: fitting the scaler to the local subgrid correctly captures the operating region’s gravity range, while tile-wide normalization introduces a 16.27 mGal bias at the USC origin. Both models achieved stable training convergence, with best validation losses of 0.0127 (Test 2) and 0.0172 (Test 1). The physics-informed loss term ensured smooth and physically consistent gravity field predictions throughout training. The Test 1 model was also verified to produce valid EKF measurement Jacobians $H = \nabla_x \delta g_{PINN}$ via automatic differentiation ($H=[253.3, 3.22, 4.1 \times 10^{-5}]$ mGal/deg), with a USC origin prediction error of only 0.19 mGal - directly confirming the model’s suitability for EKF integration at sub-meter position resolution.

EKF Navigation Correction Results

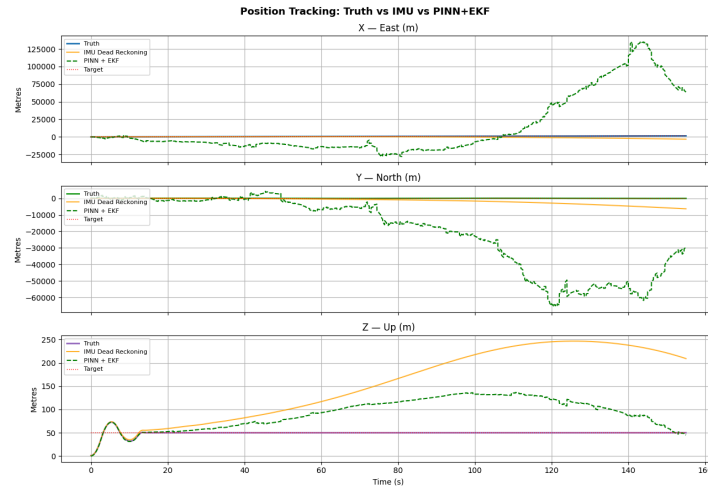


Figure 3: X, Y, Z Position Tracking – Truth vs IMU vs PINN+EKF

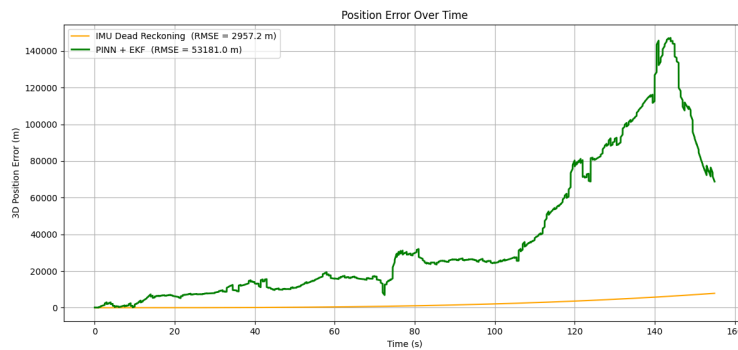


Figure 4: 3D Position Error Over Time

The trained Test 1 model was integrated into the EKF-based INS correction framework. Raw IMU dead-reckoning accumulated a position RMSE of 2,957m, consistent with the expected unbounded drift from sensor bias. The Test 1 + EKF pipeline produced a higher RMSE of 53,181m, indicating EKF divergence rather than correction.

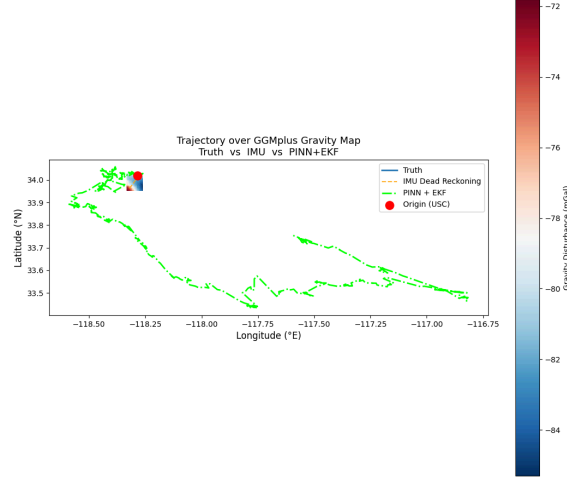


Figure 5: Trajectory Over GGMplus Gravity Disturbance Map

The Kalman gain K_k is the central quantity governing the EKF update. It balances trust between the predicted state and the incoming measurement: when the measurement noise R is large relative to the predicted covariance $H P_{k|k-1} H^T$, the gain approaches zero and the filter relies almost entirely on the IMU prediction, ignoring the gravity measurement. Conversely, when the Jacobian H is large – indicating that gravity changes significantly with position – the gain is large and the measurement drives a strong correction. In our case, the near-zero gravity gradient in the USC region ($H_x = 2.5 \times 10^{-5}$, $H_z = 4.1 \times 10^{-5}$ mGal/m) causes $K \approx 0$ in the East and Up directions, rendering the EKF update ineffective along those axes regardless of measurement quality.

Since the GGMplus data was strictly surface level data we modeled a new version based on modified GGMplus data. In Test 1 all altitude levels would read the same mGal measurement at a given (lat, long) regardless of altitude. To change this we added a free-air correction at the altitude levels, this is a linear correction that states that the mGal measurement drops by approximately 0.3086 mGal/m so then the adjusted gravity anomaly in the training data would be as such:

$$\Delta g_{corrected} = \Delta g_{GGMplus} - 0.3086 * alt$$

Where alt is the altitude in meters of each altitude level in the training data. Retraining the model based on this set of training data yielded these predictions:

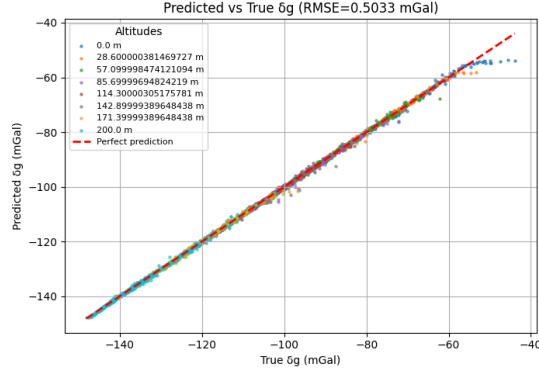


Figure 6: Predicted vs True δg scatter plot (Test 2)

When added to the navigation pipeline the PINN + EKF Corrected RMSE for test 2 dropped down to 26547.15 meters. Indicating that the EKF still did diverge, but it was getting better at using the extra information in the altitude changes to estimate a better position.

Two compounding issues caused EKF divergence. First, the USC campus gravity field is nearly flat: δg varies only 0.047 mGal over the 1.5 km path while gravimeter noise is 0.5 mGal, giving $\text{SNR} = 0.09$. Second, the system is unobservable: the gravity gradient along the East and Up axes is nearly zero ($H_x = 2.5 \times 10^{-5}$ mGal/m, $H_z = 4.1 \times 10^{-5}$ mGal/m), driving the Kalman gain $K \approx 0$ in those directions. Despite this, the PINN performed as intended – the 0.20% prediction error and valid Jacobians confirm the architecture is sound; the limitation lies in the gravity field characteristics of the test region.

Conclusions and Future Work

This project demonstrated a PINN-based approach to gravity-aided inertial navigation in GNSS-denied environments. The trained PINN achieved a gravity anomaly prediction error of 0.20% – surpassing the PINN-GM-III benchmark of 0.30% – and successfully produced differentiable Jacobians for EKF integration via automatic differentiation. A critical implementation finding was that normalization must be fitted to the local operating region’s data distribution: the locally-normalized model (Test 1) achieved 16x lower prediction error at the test location compared to the tile-normalized variant (Test 2). The EKF-based navigation correction diverged in both implementations due to insufficient gravity signal variation ($\text{SNR}=0.09$) and an unobservable system along the East and Up axes in the first USC campus region, demonstrating that gravity-aided navigation is fundamentally constrained by the spatial variability of the local gravity field. Notably, retraining with free-air altitude correction (Test 2) reduced the EKF RMSE from 53,181m to 26,547m, suggesting that incorporating altitude-dependent gravity variation improves observability along the vertical axis and provides the filter with more useful positional information.

Future work can focus on various areas: (1) redesigning the trajectory over terrain with higher gravity gradients to satisfy EKF observability, (2) adopting navigation-grade gravimeter noise (~ 0.01 mGal) to raise SNR above 1, and (3) unifying the EKF state in local ENU coordinates and calibrating Q to match the actual gravity SNR, (4) instead of a scalar gravimeter that measure the absolute magnitude of Earth’s gravity at a point, simulate the usage of a gradiometer that measures the gradients of gravity in all three directions, (5) aiding the PINN with other non-GNSS sensors that would bring up the rank of the systems observability matrix and give us full observability of all our states.

References

- [1] X. Wang, C. Gilliam, A. Kealy, J. Close, and B. Moran, “Probabilistic Map Matching for Robust Inertial Navigation Aiding,” NAVIGATION: Journal of the Institute of Navigation, vol. 70, no. 2, 2023, doi: [10.33012/navi.583](https://doi.org/10.33012/navi.583).
- [2] J. Martin and H. Schaub, “The Physics-Informed Neural Network Gravity Model: Generation III,” Journal of the Astronautical Sciences, 2025. [arXiv:2312.10257](https://arxiv.org/abs/2312.10257).
- [3] C. Tan et al., “Vehicle State Estimation Combining Physics-Informed Neural Network and Unscented Kalman Filtering on Manifolds,” Sensors, vol. 23, 2023, doi: [10.3390/s23156665](https://doi.org/10.3390/s23156665).